SE LAB 7
SHAURYA SINGH - PES1UG24CS437
FUZZ TESTING

1. python library_fines.py

```
Shelf Code: GG
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU1023
Traceback (most recent call last):
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 34, in <module>
    main()
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 20, in main
    result = parse_member_id(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 2, in parse_member_id
    member_type, number = member_id.split('-')
    ^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

```
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU-1023
Member Type and Number: ('STU', 1023)

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): 10,abc,30
Traceback (most recent call last):
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 34, in <module>
    main()
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 25, in main
    result = total_fine(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 6, in total_fine
    days = [int(d) for d in days_csv.split(',')]
            ^^^^^^
ValueError: invalid literal for int() with base 10: 'abc'
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU-1023
Member Type and Number: ('STU', 1023)

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): 10,20,30
Total Fine: 44

==== Running shelf_code ====
Enter a book title with at least two words: Dune
Traceback (most recent call last):
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 34, in <module>
    main()
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 30, in main
    result = shelf_code(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 15, in shelf_code
    return (words[0][0] + words[1][0]).upper()
                          ~~~~~^^^
IndexError: list index out of range
```

2. python -m pytest test_library_fines.py

```
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python -m pytest test_library_fines.py
============================================= test session starts =============================================
platform win32 -- Python 3.12.4, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_library_fines.py FFF                                                                            [100%]
```

```
================================================== FAILURES ==================================================
_____________________________________ test_parse_member_id_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 5, in test_parse_member_id_no_crash
  |     def test_parse_member_id_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 7, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 3, in parse_member_id
    |     return member_type.upper(), int(number)
    |                                 ^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 9, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_member_id crashed with: ValueError("invalid literal for int() with base 10: ''") for input '-'
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
```

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    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 7, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 2, in parse_member_id
    |     member_type, number = member_id.split('-')
    |     ^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 9, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_member_id crashed with: ValueError('not enough values to unpack (expected 2, got 1)') for input ''
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 7, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 2, in parse_member_id
    |     member_type, number = member_id.split('-')
    |                           ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 9, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_member_id crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s=None,  # or any other generated value
    | )
```

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_____________________________________________ test_total_fine_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 12, in test_total_fine_no_crash
  |     def test_total_fine_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 14, in test_total_fine_no_crash
    |     total_fine(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 6, in total_fine
    |     days = [int(d) for d in days_csv.split(',')]
    |             ^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 16, in test_total_fine_no_crash
    |     assert False, f"total_fine crashed with: {e!r} for input {s!r}"
    | AssertionError: total_fine crashed with: ValueError("invalid literal for int() with base 10: ''") for input ''
    | assert False
    | Failing test case: test_total_fine_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py:15
```

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    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 14, in test_total_fine_no_crash
    |     total_fine(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 6, in total_fine
    |     days = [int(d) for d in days_csv.split(',')]
    |                            ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 16, in test_total_fine_no_crash
    |     assert False, f"total_fine crashed with: {e!r} for input {s!r}"
    | AssertionError: total_fine crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_total_fine_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py:15
    +------------------------------------
```

```
_____________________________________________ test_shelf_code_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 19, in test_shelf_code_no_crash
  |     def test_shelf_code_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 21, in test_shelf_code_no_crash
    |     shelf_code(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 15, in shelf_code
    |     return (words[0][0] + words[1][0]).upper()
    |             ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 23, in test_shelf_code_no_crash
    |     assert False, f"shelf_code crashed with: {e!r} for input {s!r}"
    | AssertionError: shelf_code crashed with: IndexError('list index out of range') for input ''
    | assert False
    | Failing test case: test_shelf_code_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py:22
    +---------------- 2 ----------------
```

```
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 21, in test_shelf_code_no_crash
    |     shelf_code(s)
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\library_fines.py", line 14, in shelf_code
    |     words = title.split()
    |             ^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py", line 23, in test_shelf_code_no_crash
    |     assert False, f"shelf_code crashed with: {e!r} for input {s!r}"
    | AssertionError: shelf_code crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_shelf_code_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7\test_library_fines.py:22
    +------------------------------------
=========================================== short test summary info ============================================
FAILED test_library_fines.py::test_parse_member_id_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_library_fines.py::test_total_fine_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_library_fines.py::test_shelf_code_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================== 3 failed in 8.74s ===============================================
```

3. python library_fines.py (After fixing library_fines.py)

```
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU1023
Member Type and Number: None

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): 10,abc,30
Total Fine: 32

==== Running shelf_code ====
Enter a book title with at least two words: Dune
Shelf Code:
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU-6741
Member Type and Number: ('STU', 6741)

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): 10,20,30,40
Total Fine: 96

==== Running shelf_code ====
Enter a book title with at least two words: Gone Girl
Shelf Code: GG
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7>
```

4. python -m pytest test_library_fines.py (After fixing library_fines.py)

```
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7> python -m pytest test_library_fines.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.4, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_library_fines.py ...                                                                                                          [100%]

============================================================= 3 passed in 1.44s =============================================================
(.venv) PS C:\Users\Shaurya\Desktop\code-coverage-lab6-pes1ug24cs437\Lab 7>
```